

I. APPENDIX

A. Appendix A: The Proof of Theorem 1

Theorem 1. Let Z_{ce} be the feature representation learned by minimizing only the cross-entropy loss, and Z_{sieve} be the representation learned by Sieve with a self-supervised contrastive loss weight $\lambda > 0$. Under the idealized Information Bottleneck framework, as the compression coefficient $\beta \rightarrow \infty$, Z_{ce} converges to a minimal sufficient statistic of Y , resulting in the asymptotic loss of detection-relevant information, i.e., $I(Z_{ce}; T|Y) = 0$. Conversely, under the assumption of non-vanishing structural information, the optimization objective of Sieve provides a theoretical basis for a strictly positive lower bound for detection information, i.e., $I(Z_{sieve}; T|Y) > 0$.

Proof. Assumptions: This proof is predicated on the following theoretical assumptions: (i) the latent feature extraction follows the Markov chains $Y \rightarrow X \rightarrow Z$ and $T \rightarrow X \rightarrow Z$; (ii) the non-semantic cues T are contained within the structural information of X that is independent of Y , i.e., $I(X; T|Y) > 0$; and (iii) the model has sufficient capacity to reach the global optima of the defined objectives.

1. Asymptotic Analysis of Pure Classification (Z_{ce})

Our goal is to mathematically demonstrate that when optimizing solely for classification performance and model compression, non-semantic details in the features (captured by $X|Y$) are theoretically forced to zero.

The training process of deep neural networks can be viewed as optimizing the Information Bottleneck objective: maximizing predictive power while minimizing feature complexity. Its Lagrangian form is given by:

$$\min_{\theta} \mathcal{L}_{IB} = -I(Z; Y) + \beta I(Z; X), \quad (1)$$

where β is the Lagrange multiplier. As $\beta \rightarrow \infty$ (pursuing extreme feature compression while maintaining classification accuracy), the model tends to learn the **Minimal Sufficient Statistic** of Y [1], denoted as Z_{ce} .

By definition, Z_{ce} must satisfy two conditions simultaneously:

- 1) **Sufficiency:** $I(Z_{ce}; Y) = I(X; Y)$.
- 2) **Minimality:** $Z_{ce} = \arg \min_Z I(Z; X)$ s.t. $I(Z; Y) = I(X; Y)$.

Using the Chain Rule, we decompose $I(Z; X)$ into $I(Z; Y) + I(Z; X|Y)$ [1]. Substituting this into the minimality objective:

$$\begin{aligned} \min_Z I(Z; X) &= \min_Z [I(Z; Y) + I(Z; X|Y)] \\ &= I(X; Y) + \min_Z I(Z; X|Y), \end{aligned} \quad (2)$$

where $I(Z; Y)$ is fixed. Since mutual information is non-negative, the asymptotic solution is:

$$\lim_{\beta \rightarrow \infty} I(Z_{ce}; X|Y) = 0. \quad (3)$$

Mathematical Derivation of Detection Failure: The detection task distinguishes samples via label T , which is not fully explained by Y . Under the conditional Data Processing Inequality (DPI):

$$I(Z; T|Y) \leq I(Z; X|Y). \quad (4)$$

Combining $\lim I(Z_{ce}; X|Y) = 0$ with the non-negativity of mutual information and the Squeeze Theorem, we necessarily obtain $I(Z_{ce}; T|Y) = 0$.

2. Theoretical Characterization of Sieve (Z_{sieve})

The total loss of Sieve is $\mathcal{L}_{total} = \mathcal{L}_{ce} + \lambda \mathcal{L}_{self}$. As proven by [2], minimizing $\mathcal{L}_{self}$ is equivalent to maximizing the lower bound of $I(Z; X)$:

$$\mathcal{L}_{self} \geq \text{const} - I(Z; X). \quad (5)$$

Substituting the chain rule, Sieve's objective expands to $(1 + \lambda)I(Z; Y) + \lambda I(Z; X|Y)$.

Asymptotic Convergence Analysis:

- 1) $I(Z; Y)$ is bounded by $H(Y)$, and its gradient approaches 0 upon saturation.
- 2) Theoretically, as long as $\lambda > 0$, the optimizer attempts to maximize $I(Z; X|Y)$.

For the converged Sieve model, there must exist a constant $\epsilon > 0$ such that $I(Z_{sieve}; X|Y) \geq \epsilon$. This expands the feasible region for detection:

$$0 < I(Z_{sieve}; T|Y) \leq \epsilon. \quad (6)$$

Intuitive and Empirical Interpretation: This result implies that Sieve effectively prevents the over-compression of information that is irrelevant to the primary classification task but vital for anomaly detection. By explicitly incentivizing the retention of structural information $I(Z; X|Y)$, Sieve ensures that cues for unknown attacks (T) are preserved in the feature space, whereas they would be discarded as “noise” in a pure classification model. $\square$

B. Appendix B: The Proof of Theorem 2

Theorem 2. Let $\mathcal{Y}_s$ be the proxy target for self-supervised learning. Assuming the Label Blindness Condition $I(\mathcal{Y}_s; Y) = 0$ holds, the unsupervised feature Z_{ssl} satisfies $I(Z_{ssl}; Y) = 0$. Sieve aims to overcome this blindness condition by introducing the neighbor-consistency corrected label Y_c . Under the Manifold Smoothness Assumption and the condition of high-purity filtering, $I(Z_{sieve}; Y)$ asymptotically approaches the label entropy $H(Y)$, thus enabling the detection of adjacent unknown samples.

Proof. Formal Assumptions: 1) The input X can be decomposed into two statistically independent components X_1 and X_2 , where $\mathcal{Y}_s$ is a function of X_1 and Y is a function of X_2 . 2) The Manifold Smoothness Assumption [3] holds, meaning samples in close proximity in the feature space are highly likely to share labels. 3) The model has sufficient capacity and has converged to the global optimum of the training objective.

1. Theoretical Baseline: The Label Blindness Condition

Based on the logic of Corollary 3.3 in [1], assume $X = (X_1, X_2)$, where $X_1 \perp X_2$. X_1 determines $\mathcal{Y}_s = f_1(X_1)$, and X_2 determines $Y = f_2(X_2)$. Hence, $I(\mathcal{Y}_s; Y) = 0$. Self-supervised learning (SSL) seeks to find Z_{ssl} as the minimal sufficient statistic of $\mathcal{Y}_s$, implying $I(Z_{ssl}; X|\mathcal{Y}_s) = 0$. By the Data Processing Inequality ($Y \leftarrow X_2 \leftarrow X \rightarrow Z_{ssl}$):

$$I(Z_{ssl}; Y) \leq I(Z_{ssl}; X_2) = 0 \implies I(Z_{ssl}; Y) = 0. \quad (7)$$

Intuition: In this scenario, pure SSL is “blind” to semantic labels because it discards all information related to X_2 .

2. Mathematical Recovery of Semantic Information

Sieve filters samples using neighbor consistency $c_i \geq \xi$. According to the Manifold Smoothness Assumption [3], samples sufficiently close in feature space have a probability approaching 1 of sharing the same true label Y :

$$P(Y(x) = Y(x') \mid \|f(x) - f(x')\| < \epsilon) \rightarrow 1. \quad (8)$$

When $c_i \rightarrow 1$, the posterior probability $P(Y = Y_c) = 1 - P_e$ is maximized. In the asymptotic limit where the filtering threshold $\xi \rightarrow 1$, the error rate $P_e \rightarrow 0$. According to Fano’s Inequality:

$$H(Y|Y_c) \leq H_b(P_e) + P_e \log(|\mathcal{Y}| - 1) \xrightarrow{P_e \rightarrow 0} 0. \quad (9)$$

Therefore, $I(Y_c; Y) = H(Y) - H(Y|Y_c) \approx H(Y)$. This indicates that Y_c theoretically retains nearly all information of the true labels Y .

3. Convergence and Information Bound

Sieve minimizes $\mathcal{L}_{ce}(Z, Y_c)$, which is equivalent to maximizing the lower bound of $I(Z; Y_c)$. Assuming model convergence, $I(Z_{sieve}; Y_c) \rightarrow H(Y_c)$. Combining the mutual information expansions:

$$I(Z; Y) = I(Z; Y_c) + I(Z; Y|Y_c) - I(Z; Y_c|Y). \quad (10)$$

Under Sieve’s rigorous filtering, Y_c is almost a deterministic function of Y , making $H(Y_c|Y)$ and $I(Z; Y_c|Y)$ negligible. Since $I(Z; Y|Y_c) \geq 0$:

$$I(Z_{sieve}; Y) \geq I(Z_{sieve}; Y_c) - H(Y_c|Y) \approx H(Y). \quad (11)$$

Intuitive and Empirical Interpretation: This formal derivation provides a theoretical guarantee that $I(Z_{sieve}; Y)$ approaches the label entropy, effectively overcoming label blindness.

- *Semantic Encoding:* Unlike pure SSL, Sieve forces the encoding of X_2 , ensuring known samples x_{in} form compact clusters around semantic centers.
- *Detection Separability:* Because $Z(x_{in})$ clusters tightly while unknown samples x_{out} (which possess different X_2 features) are not pushed toward these centers, the two categories become separable in the feature space.

Thus, the theoretical recovery of semantic information is the key driver for Sieve’s superior detection performance. $\square$

C. Appendix C: The Proof of Theorem 3

Theorem 3. *Let Z follow a class-conditional distribution with covariance matrix Σ_k . The self-supervised optimization objective of Sieve ($\mathcal{L}_{self}$) is mathematically equivalent to minimizing the trace of the intra-class covariance matrix Σ_k , i.e., $\text{Tr}(\Sigma_k) \rightarrow 0$. Under the assumption of a class-conditional Gaussian distribution, this implies that the determinant $\det(\Sigma_k) \rightarrow 0$. In the asymptotic limit, this compactness causes the eigenvalue amplification factor of the Mahalanobis distance metric to tend towards infinity, thereby minimizing the theoretical bound of the Open Set Risk.*

Proof. Assumptions: The following characterization is predicated on three key theoretical assumptions: (i) the latent features Z for each known class follow a multivariate Gaussian distribution $Z|Y = k \sim \mathcal{N}(\mu_k, \Sigma_k)$; (ii) the contrastive alignment loss $\mathcal{L}_{align}$ is optimized to its global minimum; and (iii) sample pairs are drawn independently from the underlying distribution.

To quantify the safety of the detector in unknown environments, we cite the definition from [4].

Definition: The Open Set Risk $R_{\mathcal{O}}$ is defined as the effective measure (Lebesgue measure) of the recognition function f over the open space $\mathcal{O}$ (the space outside the support of the training data):

$$R_{\mathcal{O}}(f) = \frac{\int_{\mathcal{O}} f(z) dz}{\int_{S_o} f(z) dz} \quad (12)$$

- $f(z)$ is the recognition function: if z is classified as a known class (ID), then $f(z) = 1$, otherwise 0.
- S_o represents the entire feature space.

Minimizing $R_{\mathcal{O}}$ necessitates minimizing the volume of the acceptance region of $f(z) = 1$ within the feature space. Theoretically, the more compact the distribution of known classes, the lower the probability of erroneously infringing upon the open space $\mathcal{O}$.

1. Mathematical Derivation of Covariance Compression

Under the Gaussian assumption, the differential entropy is given by:

$$H(Z|Y = k) = \frac{1}{2} \ln((2\pi e)^d \det(\Sigma_k)), \quad (13)$$

where $\det(\Sigma_k)$ is geometrically proportional to the square of the volume of the distribution ellipsoid:

$$\text{Vol}(\text{ID Distribution}) \propto \sqrt{\det(\Sigma_k)}. \quad (14)$$

Thus, within this framework, minimizing $H(Z|Y)$ is equivalent to minimizing $\det(\Sigma_k)$.

Citing [5], the $\mathcal{L}_{self}$ loss asymptotically optimizes Alignment and Uniformity. For positive sample pairs belonging to the same class k , the Alignment loss is $\mathcal{L}_{align} \triangleq \mathbb{E}_{(z_i, z_j) \sim P_k} \|z_i - z_j\|^2$. Expanding this under the independent sampling assumption:

$$\mathbb{E}\|z_i - z_j\|^2 = \text{Tr}(\Sigma_k) + \text{Tr}(\Sigma_k) - 0 = 2\text{Tr}(\Sigma_k), \quad (15)$$

where $\text{Tr}(\Sigma_k)$ is the sum of eigenvalues. By minimizing $\mathcal{L}_{self}$, Sieve directly forces $\text{Tr}(\Sigma_k) \rightarrow 0$. By the AM-GM inequality, $\det(\Sigma_k) \leq (\text{Tr}(\Sigma_k)/d)^d$. By the Squeeze Theorem:

$$\lim_{\text{train} \rightarrow \infty} \det(\Sigma_k) = 0. \quad (16)$$

2. Geometric and Empirical Interpretation

Geometrically, the collapse of $\det(\Sigma_k)$ creates a “potential well” for ID samples. Sieve employs the Mahalanobis distance:

$$D_M(z) = \sqrt{(z - \mu_k)^T \Sigma_k^{-1} (z - \mu_k)}. \quad (17)$$

As $\text{Tr}(\Sigma_k) \rightarrow 0$, eigenvalues $\lambda_i \rightarrow 0^+$, and the reciprocal eigenvalues in Σ_k^{-1} tend to infinity. For any unknown sample $z_{out} \neq \mu_k$, the squared distance $D_M^2(z_{out}) \geq v_j^2/\lambda_j \rightarrow \infty$.

3. Convergence of Open Set Risk

The volume of the acceptance region is $V(A_\tau) \propto \sqrt{\det(\Sigma_k)}$. Combining $\det(\Sigma_k) \rightarrow 0$ with the definition from [4]:

$$\lim_{\text{train} \rightarrow \infty} V(A_\tau) = 0. \quad (18)$$

This proves that in the infinite open space $\mathcal{O}$, the overlap $\int_{\mathcal{O}} f(z) dz$ is theoretically constrained to zero under these idealized conditions.

In conclusion, Sieve provides a theoretical guarantee for minimizing Open Set Risk by forcing feature distribution collapse. While the empirical risk depends on the degree to which features adhere to the Gaussian distribution, this optimization significantly maximizes the separability of known and unknown traffic in the distance space. $\square$

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